

Build the Recommender

Project: Transparent Book Recommender

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Project Problem and Success Test

Without opening the old lesson, explain one key idea from Project Problem and Success Test. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Agents and Tools

Without opening the old lesson, explain one key idea from Agents and Tools. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Predictive vs Generative

Without opening the old lesson, explain one key idea from Predictive vs Generative. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Handle no match for Implementing a recommender. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Implementing a recommender. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Changed independent case

Request: mystery, at most 100 pages. Catalogue: D mystery110 illustrated; E mystery80 plain; F science80 illustrated; G mystery95 illustrated; H mystery90 illustrated. Filter first; score 3 for mystery plus 2 for illustrations; break ties by fewer pages.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. Your learning target

Catalogue: A mystery/90/illustrated, B mystery/140/not illustrated, C science/80/illustrated. Request mystery, maximum 100 pages. Rank with 2 genre points and 1 illustration point after filtering. Show why other books are excluded.

6. Solve the given case

A has flags (1,1), B=(1,0), C=(0,1); score=2g+l.

Compute the winner and evidence to display.

Show your trace, calculation or evidence

Write or draw your reasoning and evidence.

7. Check a changed case

A reader requests space and at most 60 pages. Book A is space, 90 pages; B is animals, 40 pages. What should the prototype return if both conditions are required?

My independent answer and reason

Write or draw your reasoning and evidence.

Week 33 Test and reflect

8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record · My evidence

Test record: Changed input or condition

My evidence: _____

Test record: Expected answer or fallback

My evidence: _____

Test record: Observed result

My evidence: _____

Test record: Why the result occurred

My evidence: _____

9. Project checkpoint

This is the complete baseline: inputs, scoring, filter, ranking, explanation, and abstain rule.

The project change and evidence I will keep

Write or draw your reasoning and evidence.

10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

11. Transfer practice from the original programme

Catalogue: A mystery/90/illustrated, B mystery/140/not illustrated, C science/80/illustrated. Request mystery, maximum 100 pages. Rank with 2 genre points and 1 illustration point after filtering. Show why other books are excluded.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

12. W1 Trace selection

Why does A win the worked example even though B is processed first?

My reasoning and evidence

13. W2 Handle no match

State `recommend([], 10)` and the worked items with limit 8.

My reasoning and evidence

14. W3 Repair order dependence

A tied selection changes after reversing input order. Identify the likely missing policy and test.

My reasoning and evidence

15. W4 Implement independently

Use the function with X (fun 4, learning 1, minutes 12), Y (3,3,10), Z (5,5,13), limit 12. Predict the winner, reverse the list and retest. Also test limit 9.

My reasoning and evidence

16. W5 Explain the result

Write a one-sentence explanation of X winning without calling it objectively best.

My reasoning and evidence

17. Building the data pipeline — W1 Trace a validation

Explain why a row with fun "4" fails while fun 4 can pass.

My reasoning and evidence

18. Building the data pipeline — W2 Check missing fields

What does `valid_item` return if learning is absent?

My reasoning and evidence

19. Building the data pipeline — W3 Repair the connection

A program validates rows but still sends the original raw list to recommend. What should change?

My reasoning and evidence

20. Building the data pipeline — W4 Build independently

Combine validation and the Week 24 selector for A (minutes 10, fun 3, learning 2), B (12,4,1), and C (8,"5",5), limit 10. State accepted rows, rejected row, selection and explanation.

My reasoning and evidence

21. Building the data pipeline — W5 Handle duplicates

Two valid rows have the same name but different ratings. Should individual validation alone permit selection?

My reasoning and evidence

22. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

23. Pictorial case practice: Build the Recommender

A fictional book chooser filters a request before comparing preference scores.

Request: animal subject, at most 90 pages. Score 2 for illustrated and 1 for comic. Tie chooses smaller ID.

A animal90 illustrated prose; B animal70 plain comic; C space60 illustrated comic; D animal80 illustrated comic.

Filtering before scoring

	Pages	Preference score	Status
A	90	2	eligible
B	70	1	eligible
C	60	3	subject fails
D	80	3	eligible

Read a cell using both its row and column labels.

Build the Recommender

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Filter

A,B,D meet subject and page requirements; C fails subject.



2. Compute scores

A=2; B=1; D=3.



3. Choose best

Return D with both preference matches recorded.



4. Deterministic output

Reordering the input rows must not change D's unique winning score.

CHANGE ONE THING

Change D's pages to 91, leaving every other field unchanged.

Explain the changed result using the matching lesson.

Explain why B does not win after D is excluded even though it has fewer pages.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: Save the recommender code, contract and this full trace; test the same records in different input orders.

24. Project portfolio milestone

Project: Transparent Book Recommender

Current milestone: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with evidence.

Save the recommender code, contract and this full trace; test the same records in different input orders.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

25. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional book chooser filters a request before comparing preference scores.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Reordering the input rows must not change D's unique winning score.

1. Choose best
2. Filter
3. Deterministic output
4. Compute scores

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Deterministic output Reordering the input rows must not change D's unique winning score. Eligibility and ranking answer different questions. C has a high preference score but is not an animal book, so it must stay excluded. The implementation needs a no-match result and explicit tie policy even when this example has a unique winner. Save rejected reasons for an explanation, not just the final ID.

Practice 2: explain the deciding evidence

Changed-case practice: Change D's pages to 91, leaving every other field unchanged. Which conclusion is supported by the supplied facts?

1. D becomes ineligible, so A wins with score 2.
2. This one example guarantees correct results for every future input.
3. We can decide the result without examining the changed input or the procedure.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: D becomes ineligible, so A wins with score 2. The strict excess over the 90-page maximum changes filtering before ranking.

Repair a missing building block

Review Project Problem and Success Test (ai-2/32). Goal: Define a small application purpose. Short practice: Write an equality test and explain which requirement it checks.. Return to this week and explain what you changed.

Review Agents and Tools (ai-2/30). Goal: Explain the mechanism and terminology of state machines for bounded agents. Short practice: Can the assistant accept Trace B because its final state is done?. Return to this week and explain what you changed.

Review Predictive vs Generative (ai-2/27). Goal: Distinguish discriminative prediction from generation by inspecting the requested output. Short practice: Does matching the physical box in the changed table prove the generator measures boxes?. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual output or error, change one input, then run again. Keep code, predictions, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

